

Information-Geometric Optimization of Fokker-Planck Dynamics in Neuromorphic Synthetic Synapses: A Framework for Near-Equilibrium Distribution Learning

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April 25, 2026

Abstract

This document outlines a proposed research direction at the intersection of information geometry, stochastic dynamics, neuromorphic engineering, and synthetic biology. The central hypothesis is that biological synapses learn efficiently by operating near thermodynamic equilibrium, and that this principle can be formalized using the Fokker-Planck (FP) equation and the Fisher information metric. I seek mentorship to transform this theoretical framework into a computational proof-of-concept suitable for a future PhD application.

1 Motivation and Three Core Questions

The human brain computes with 20 watts, while state-of-the-art AI consumes megawatts. Bridging this gap requires understanding how biological systems learn with extreme energy efficiency. My work is guided by three interconnected questions:

1. **Information geometry and metabolic cost:** Can we prove that the brain minimizes Fisher information along its learning trajectories to use the least metabolic energy? This would unify the Free Energy Principle with information-theoretic efficiency.
2. **Diffusion as a natural regularizer:** How does the diffusion term in the FP equation prevent catastrophic forgetting in hybrid AI systems? I hypothesize that intrinsic noise acts as an implicit Bayesian regularizer, analogous to natural gradient methods for continual learning.
3. **Near-equilibrium computation:** By constraining FP dynamics to near-equilibrium states, can we achieve the same computational density as the human brain on silicon? This question directly informs the design of energy-efficient neuromorphic hardware.

2 Proposed Technical Approach

2.1 Fokker-Planck for Neural Populations

The FP equation describes the time evolution of the probability density $p(v, t)$ of membrane potentials in a population of stochastic leaky integrate-and-fire neurons:

$$\frac{\partial p(v, t)}{\partial t} = -\frac{\partial}{\partial v} [\mu(v, t)p(v, t)] + \frac{\sigma^2}{2} \frac{\partial^2 p(v, t)}{\partial v^2}, \quad (1)$$

where $\mu(v, t)$ is drift (deterministic input) and σ is diffusion (noise). My initial goal is to numerically solve this equation using a shape-morphing Gaussians method (2023), which automatically preserves probability mass.

2.2 Fisher Information as a Learning Metric

For a parametric family of probability distributions $\{p(x; \theta)\}$, the Fisher information metric is:

$$\mathcal{I}_{ij}(\theta) = \mathbb{E} \left[\frac{\partial \log p}{\partial \theta_i} \frac{\partial \log p}{\partial \theta_j} \right]. \quad (2)$$

I hypothesize that learning trajectories that minimize the path length in Fisher space (the *Fisher length*) correspond to maximally energy-efficient updates. This is mathematically equivalent to natural gradient descent:

$$\tilde{\nabla} L(\theta) = \mathcal{I}(\theta)^{-1} \nabla L(\theta). \quad (3)$$

2.3 Mapping to Synthetic Synapses

Recent advances in DNA-based memristors (Keremane & Poudel, Penn State) and droplet membrane ion channel synapses (Sarkar lab, MIT) offer physical substrates for implementing near-equilibrium learning. The FP diffusion term σ can be tuned via ion channel density or compliance current. I plan to simulate how varying σ affects learning stability and energy consumption.

3 Current Status and Immediate Needs

I am a third-year biochemistry student with basic programming knowledge (Python, C, Linux, $\text{\LaTeX}$). I have begun implementing a deterministic LIF neuron simulator as a foundation. However, I lack mentorship in:

- Numerical methods for solving the FP equation efficiently.
- Connecting information geometry to concrete learning algorithms.
- Designing simulations that produce publishable results.

4 Request for Mentorship

I am seeking a 6–12 month remote mentorship relationship with a professor or senior PhD student. Specifically, I would benefit from:

1. Guidance on which numerical libraries (e.g., ‘pyddm’, ‘JAX’, ‘BrainPy’) are most appropriate.
2. Feedback on a 2–3 page research statement to be included in PhD applications (targeting Stanford, MIT, Harvard, Princeton).

5 Long-Term Vision

If mentored successfully, I aim to produce a preprint on bioRxiv by the end of my fourth year, titled “*Fisher Information-Guided Learning in Near-Equilibrium Neuromorphic Synapses*”. This work would serve as the foundation for a PhD thesis at a top-tier US institution, ideally in a lab combining theoretical neuroscience with neuromorphic engineering.

Acknowledgments

I thank you for considering this request. Any guidance, no matter how brief, will be deeply appreciated.

References

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